House Price Prediction using Linear Regression

1. Introduction

Machine Learning is a powerful technology that allows computers to learn from data and make predictions without being explicitly programmed.

In this project, we build a Linear Regression model to predict house prices using the California Housing dataset.

The main goal of this project is to understand the complete Machine Learning workflow including data loading, preprocessing, model training, and evaluation.

2. Dataset Description

We used the California Housing dataset, which contains information about houses in different districts of California.

Features (Input Variables):

- **MedInc** – Median income of residents

- **HouseAge** – Average age of houses

- **AveRooms** – Average number of rooms

- **AveBedrms** – Average number of bedrooms

- **Population** – Population in the area

- **AveOccup** – Average number of people per house

- **Latitude & Longitude** – Location of houses

Target Variable:

- **MedHouseVal** – Median house value (price)

3. Exploratory Data Analysis (EDA)

Before training the model, we analyzed the dataset:

- Used ".info()" to check data types

- Used ".describe()" for statistical summary

- Checked for missing values using ".isnull()"

- Plotted histograms to understand data distribution

Observation:

The dataset is clean and does not contain missing values.

All features are numerical, so it is suitable for Linear Regression.

4. Model Used: Linear Regression

We used the Linear Regression algorithm, which is one of the simplest and most commonly used machine learning models.

Working Principle:

Linear Regression finds a relationship between input variables (X) and output (Y) using a straight line:

[Y = mX + c]

It tries to minimize the error between actual and predicted values.

5. Model Training

- The dataset was split into:
  - **80% Training data**
  - **20% Testing data**
- The model was trained using the training data
- Predictions were made using the testing data

6. Evaluation Metrics

To evaluate the model performance, we used:

1. Mean Absolute Error (MAE):

It calculates the average difference between actual and predicted values.

2. Root Mean Squared Error (RMSE):

It gives more weight to large errors.

3. R^2 Score:

It shows how well the model fits the data (value between 0 and 1).

Results:

- MAE: 0.533

- RMSE: 0.746

- R^2 Score: 0.576

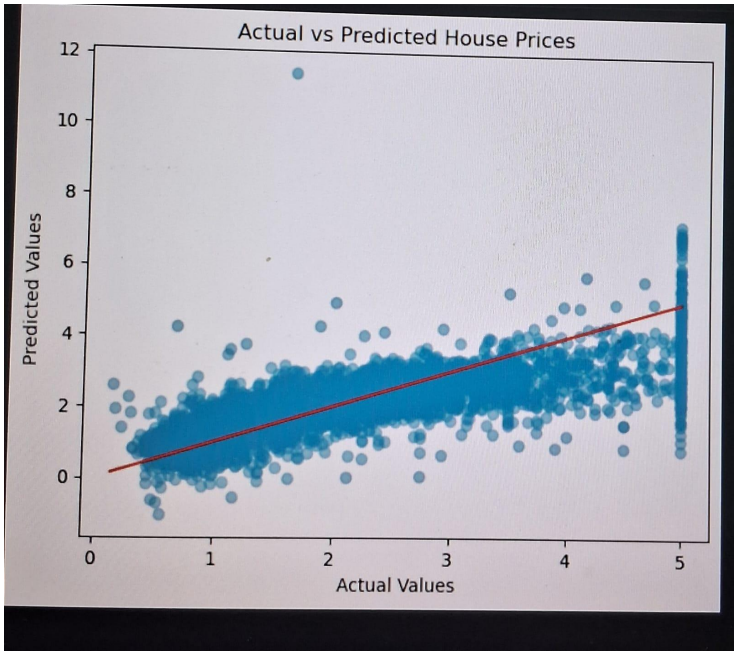
7. Results and Visualization

- A scatter plot was created between Actual vs Predicted values

- A red line shows ideal predictions

Observation:

Most data points are close to the line, which means the model is performing reasonably well.



8. Conclusion

In this project, we successfully built a Linear Regression model to predict house prices.

The model gives good results, but it is not perfect.

This project helped us understand the complete machine learning workflow in a simple and practical way.

9. Future Improvements

- Use advanced models like Decision Tree or Random Forest
- Perform feature engineering
- Tune model parameters for better accuracy
- Handle outliers for improved performance

10. Tools & Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Thank You